

Software Engineer with a degree from the Higher School of Communication of Tunis (SUP'COM). I specialize in embedded Linux C/C++ applications for payment terminal systems, with hands-on experience in protocol implementation, terminal certification, and Linux-embedded environments. A strong algorithmic foundation from competitive programming, including the Africa and Arab Collegiate Programming Championship (ACPC), complements my problem-solving approach to software engineering.

TECHNICAL EXPERIENCE

C/C++ Software Engineer

September 2024 — Present

Telnet / Ingenico Partner

Sfax, Tunisia

- Developed payment features for embedded terminals, including NFC, contact, and magstripe transaction support.
- Updated and maintained existing implementations, resolving field issues on deployed payment terminals.
- Analyzed and resolved issues raised during Self 2000 LE terminal certification by a certification lab (Pagateq).
- Developed, enhanced, and maintained C/C++ applications for embedded payment terminals and monetics solutions in collaboration with Ingenico.
- Worked in a Linux embedded environment, optimizing software performance and stability.
- Used Git and SVN for version control, Jira for task tracking, and worked in a Docker-based development environment.

Freelance Developer

January 2023 - September 2024

Upwork

Remote

- Analyzed client requirements and converted them into functional, live applications.
- Developed and maintained applications using Python, Java, and C/C++.
- Implemented new features and optimized existing code to improve performance and usability.

Backend Dev / End-of-Studies Project: Containerized web app for web analysis

May 2023 – September 2023

Niiwaa

Tunis, Tunisia

- Designed the architecture of the MongoDB database, meticulously structuring it to efficiently handle the extensive volume of collected data.
- Architected the project, carefully delineating the master and worker architectures, and selecting optimal technologies.
- Developed the backend of the application using Express.js, ensuring robust performance and reliability in data extraction and processing.
- Implemented the master-slave architecture, orchestrating the connection between workers and the master node using RabbitMQ and Celery.
- Containerized the entire application using Docker, encapsulating each component within its own container for efficient deployment and management.
- Orchestrated the application using Docker Compose, enabling seamless management of the entire web analysis platform.

EDUCATION

ICT engineering, Higher School Of Communication Of Tunis

2019 - 2023

Pre-Engineering, Preparatory Institute for Engineering Studies of Nabeul (rank 11/700 at National Exam)

2017 - 2019

Baccalaureate degree with Excellent grade (16.59/20), Benguerden's High School

2013 - 2017

PROJECTS

Open-CV Virtual Mouse (Python) : Webcam hand-tracking to simulate mouse movement and clicks.

October 2021

Deep RL Snake Game Player (Python) : Trained an RL agent to play Snake via reward feedback.

December 2021

Basic Routing Algorithm (C++) : Link-state & distance-vector routing using Dijkstra's algorithm.

December 2021

Path Finding Visualizer (ReactJS) : Visualizes pathfinding algorithms solving maze routes.

November 2023

Data Compressor (C) : File compression tool implementing Huffman encoding to reduce file size.

February 2025

SKILLS

Tools and Languages

Python, C/C++, Rust, JavaScript, Express.js, ReactJs, Docker, Git, SVN

Embedded & Payments

Embedded Linux, ZVT/OPI/ECR Protocols, EMV Certification, P2PE, HSM

Core Competencies

Algorithmic Problem Solving, Mathematical Optimization

Communication

English (Fluent), French (Fluent), Arabic (Native), German (Basics)

ACTIVITIES

Started my own e-commerce business

April 2022 - 2023

Trained high school students for IOI contest

Mars 2022

Problem solving contestant : Honorable mention in ACM Arab Collegiate Programming Contest 2021

Mars 2021

Participated in a rescue workshop with the Red Crescent and received training

August 2019